and one if $z \in \Sigma_E \setminus \mathcal{A}$ (the symbol $\setminus$ means the removal of the second set from the first). Then for any trajectory starting in $\mathcal{A}$, we have that the average of χ along Γ vanishes, i.e.

$$\bar{\chi}_\Gamma = \lim_{\tau\to\infty} \tau^{-1} \int_0^\tau \chi(z(t))\mathrm{d}t = 0,$$

since $\mathcal{A}$ is an invariant set.

On the other hand, the average of χ on Σ_E is just the probability measure of $\Sigma_E \setminus \mathcal{A}$. If we assume the system is ergodic, we must have $\Pr(\Sigma_E \setminus \mathcal{A}) = 0$, thus $\Pr(\mathcal{A}) = 1$. Since, according to the definition, equality of time and space averages is allowed to fail only on a set of probability measure zero, we have that either $\Pr(\mathcal{A}) = 0$ or $\Pr(\mathcal{A}) = 1$.

If a system is not ergodic, then there will be an invariant set $\mathcal{A}$ whose probability measure is strictly between zero and one. The trajectories emanating from this set are unable to explore the entire energy surface and the corresponding trajectory averages will, at least for some observables, be different than the spatial average of that observable. A system which is not microcanonically ergodic has an energy surface that can be partitioned into subdomains of nonzero measure in which trajectories are forever trapped.

Often, the strict definition of ergodicity must be relaxed to accommodate the realistic modelling setting; for example only some portion of the energy surface may be accessible by trajectories in the time-interval available, due to the cost of computing steps with a numerical method. For a more detailed discussion of these and other ergodicity issues in the context of deterministic molecular dynamics, see the articles of Tupper [379, 381].

Given a system with energy of the form $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$, what can be said about the ergodicity of the system on the energy surface $H = \mathrm{E}$? Coverage of the set may follow in certain cases from periodic or quasiperiodic motion in which the trajectories wrap around the surface filling in space.

As a very simple example, the harmonic oscillator is always ergodic on its energy surfaces (i.e. energy curves, in this case). Some examples of two-dimensional systems are shown in Fig. 5.3.

Ergodicity is a concept that often becomes more relevant in systems with many degrees of freedom. For example if one thinks of having not just one but many double well oscillators connected together in some way so that they can exchange energy, then it is likely that the energy can accumulate, given enough time, in one of the oscillators, enabling it to overcome the barrier.

Example 5.4 (2D Harmonic System) Consider a two-dimensional harmonic system, i.e. the system with Hamiltonian

$$H = p_1^2/2 + p_2^2/2 + \Omega_1^2 q_1^2/2 + \Omega_2^2 q_2^2/2,$$

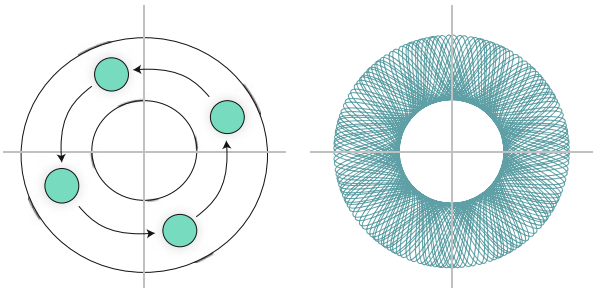


Fig. 5.3 Two examples in which we suppose the accessible domain at given energy is the annulus. On the *left*, periodic transport of a collection of points with positive area is evidence of non-ergodic motion (the union of the four disks shown would constitute an invariant region of positive measure). In the figure at *right* a quasi-periodic space-filling trajectory is shown. The oscillatory motion, although quite regular in appearance, gradually fills in the whole domain

where Ω_1, Ω_2 are two nonzero parameters. Because the two systems involving q_1, p_1 and q_2, p_2 are not coupled in the equations of motion, this system has two invariants $H_1 = p_1^2/2 + \Omega_1^2 q_1^2/2$ and $H_2 = p_2^2/2 + \Omega_2^2 q_2^2/2$. For any $\mathrm{E}_1, \mathrm{E}_2 > 0$, the set of points satisfying $H_1 = \mathrm{E}_1$, $H_2 = \mathrm{E}_2$ is a two-dimensional torus. If the frequencies Ω_1 and Ω_2 are not rationally related (i.e. Ω_1/Ω_2 is an irrational number), then for any $\mathrm{E}_1, \mathrm{E}_2$ the torus will eventually be filled in by any trajectory. On the other hand, the energy surface $H = \mathrm{E}$ is the union of all such 2-tori with $\mathrm{E}_1 + \mathrm{E}_2 = \mathrm{E}$. There is no mechanism for the system to move from one torus to the other, so the system is not ergodic.

In typical molecular dynamics applications with multiple bodies and complicated force laws, the motion is assumed to be chaotic. Ergodic coverage of Ω_E would then have to arise from the global properties of a chaotic system (sensitivity to initial conditions and transitivity). Liouville's equation,

$$\frac{\partial \rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

describes how a distribution of initial conditions is carried forward under the time evolution of a dynamical system. That is, starting from the initial distribution of phase space points described by $\rho(z, 0)$, we arrive at a new distribution $\rho(z, t)$. We will see in an example to be presented below that various sets of initial conditions with uniform energy E may appear to spread out to cover (become dense in) a particular region, the accessible energy surface at energy level E.

An even stronger property than the ergodic property is the concept of a *mixing* system. For a mixing system, the finite time density $\rho(z, t)$ converges, in the weak sense, to the invariant distribution $\rho_\infty(z)$, as $t \to \infty$. That is, we have, for all test functions ϕ in some chosen space

$$\lim_{t\to\infty} \langle \phi, \rho(\cdot, t) \rangle = \langle \phi, \rho_\infty \rangle.$$